

Musculoskeletal, Connective Tissue and Integumentary System 2

MDS221 — Head, Neck & Trunk, with Muscle Physiology

QUESTIONS

131

SECTIONS

8

COVERAGE

Lectures 1-12

FORMAT

Single best answer

This is every question from the MDS221 past paper, rebuilt. The original ran to **190 items** but repeated itself heavily — the inferior oblique was asked three separate times, the inguinal region four. Those repeats are merged here, and **15 questions on the limbs, shoulder and knee were removed** because they belong to MDS220, not to any MDS221 lecture. What is left is **131 distinct questions**, each one traceable to a lecture in your syllabus.

BUILT FROM

MDS221 past paper (190 items, all recovered) · MDS221 course syllabus, Lectures 1-12 · cross-checked against the MDS212 and MDS211 recall papers for repeated examiner patterns. Questions outside the MDS221 lecture list were not carried over.

HOW THIS PAPER IS BUILT

Every answer choice is written out in full — no abbreviations, so you cannot shortcut by spotting the one spelled-out option. Choices within a question are matched to within a few characters of each other, and the correct answer is spread evenly across A to E, so neither length nor position gives anything away. Sit it closed-book, then read the key: the reasoning matters more than the letter.

01 MUSCLE PHYSIOLOGY

LECTURES 10-12

1. The force of contraction in cardiac muscle can be increased by which of the following mechanisms?

- (A) Decreasing the calcium induced calcium release
- (B) Increasing the opening time of the dihydropyridine channel
- (C) Increasing the opening time of the ryanodine receptor
- (D) Increasing the sarcoplasmic calcium pump activity
- (E) Increasing the sodium calcium exchanger activity

2. Which of the following components would you not find in a smooth muscle cell?

- (A) Calmodulin
- (B) Tropomyosin
- (C) Troponin
- (D) Gap junction
- (E) Dense body

3. In gastrointestinal smooth muscle contraction, which event occurs first?

- (A) Calcium induced calcium release
- (B) Myosin light chain becomes phosphorylated
- (C) Myosin light chain kinase becomes activated
- (D) Myosin attaches onto the actin filament
- (E) Membrane depolarisation of the cell

4. Which characteristic is not shared by skeletal, cardiac, and smooth muscle?

- (A) Calcium induced calcium release
- (B) Actin and myosin filaments
- (C) Action potential triggered contraction
- (D) Voltage gated calcium channels
- (E) Adenosine triphosphate requirement

5. In skeletal muscle, where is the transverse tubule system located?

- (A) At the junction of A and I bands
- (B) In the I band of sarcomere
- (C) At the M line of sarcomere
- (D) At the Z line of sarcomere
- (E) In the H zone of sarcomere

6. An absence of the protein dystrophin primarily disrupts which skeletal muscle component?
- (A) The sarcolemma costamere
 - (B) The transverse tubule
 - (C) The sarcoplasmic reticulum
 - (D) The myofibril bundles
 - (E) The mitochondrial membrane
7. In smooth muscle, where is calcium primarily sequestered?
- (A) Terminal cisternae only
 - (B) Sarcoplasmic reticulum
 - (C) Sarcoplasm of the cell
 - (D) Caveolae of membrane
 - (E) Mitochondrial matrix
8. Muscle atrophy from prolonged immobilisation in a cast is chiefly caused by which process?
- (A) Fibroblast rapid hyperplasia
 - (B) Motor neuron cell apoptosis
 - (C) Degradation exceeding synthesis
 - (D) Myosin light chain phosphorylation
 - (E) Satellite cell early senescence
9. Which statement about skeletal muscle structure is correct?
- (A) A triad has two tubules and one cisterna
 - (B) The sarcomere is the storage site of calcium
 - (C) Caveolae are invaginations of the sarcolemma
 - (D) The sarcolemma is the plasma cell membrane
 - (E) The sarcoplasmic reticulum lies at the Z line
10. The activity of the sarcoplasmic reticulum calcium pump in cardiac muscle is regulated by which protein?
- (A) Myosin light chain kinase
 - (B) Phospholamban protein
 - (C) Inositol trisphosphate
 - (D) Cyclic guanosine monophosphate
 - (E) Phospholipase C enzyme
11. Multi-unit smooth muscle cells are characterised by which feature?
- (A) The cells are each innervated individually
 - (B) The cells are coupled by many gap junctions
 - (C) The cells are found mainly in the oesophagus
 - (D) The cells are joined together by desmosomes
 - (E) The action potential spreads from cell to cell
12. In skeletal muscle, which event occurs before transverse tubule depolarisation?
- (A) Calcium is taken back into store
 - (B) Calcium release channels open up
 - (C) Calcium binds onto troponin C
 - (D) Sodium channels of membrane open
 - (E) Actin binds onto the myosin head
13. What causes rigor mortis in skeletal muscle?
- (A) A lack of action potential firing
 - (B) An excess of adenosine triphosphate
 - (C) A fall of intracellular calcium
 - (D) A lack of adenosine triphosphate
 - (E) A rise of intracellular calcium
14. Which characteristic is not shared by skeletal and cardiac muscle?
- (A) Troponin complex of filament
 - (B) Action potential triggers entry
 - (C) Calcium rise needed for force
 - (D) Actin thin filament of muscle
 - (E) Calcium induced calcium release

15. In gastrointestinal smooth muscle, which event immediately follows the binding of calcium to calmodulin?

- (A) Calcium induced calcium release begins
- (B) Membrane depolarisation is initiated
- (C) Ligand gated calcium channels open
- (D) Myosin light chain kinase is activated
- (E) Intracellular calcium rises further

16. How can cardiac muscle contractile force be increased?

- (A) By decreasing the phospholamban activity
- (B) By shortening the dihydropyridine time
- (C) By increasing sodium calcium exchange
- (D) By blocking the ryanodine receptor channel
- (E) By decreasing calcium induced release

17. Which statement about the sarcomere is correct?

- (A) Actin binds the nucleotide at active site
- (B) The thin filament forms the whole A band
- (C) The M line anchors the thin filament ends
- (D) The thick filament forms the whole H band
- (E) The Z line anchors the thin filament ends

18. Type I and Type IIb muscle fibres are primarily classified based on which property?

- (A) Myosin adenosine triphosphatase activity
- (B) Myoglobin content within the fibre
- (C) Capillary density around the fibre
- (D) Glycogen content within the fibre
- (E) Muscle fibre cross sectional diameter

19. If the calf muscle is immobilised in a cast for a long time, what change would you expect?

- (A) Fibres convert to type one
- (B) Muscle mass is lost slowly
- (C) Protein synthesis rate rises
- (D) Muscle fibre number rises
- (E) Muscle degeneration lessens

20. Which of these is the best example of an isometric muscle contraction?

- (A) Squeezing buttock without moving
- (B) Lowering a dumbbell down slowly
- (C) Jumping vertically off the ground
- (D) Walking up a steep hill slope
- (E) Rising up from a low chair seat

21. During a graded muscle contraction, which motor units are typically recruited first?

- (A) Small units resistant to fatigue
- (B) Units of highest firing threshold
- (C) Random units of every fibre type
- (D) Large units of fast twitch fibre
- (E) Units of type two b muscle fibre

22. Cardiac glycoside drugs increase the contractility of the heart primarily by which effect?

- (A) Increasing the calcium pump activity
- (B) Opening the ryanodine receptor gate
- (C) Blocking the L type calcium channel
- (D) Activating cyclic guanosine monophosphate
- (E) Inhibiting the sodium potassium pump

23. In pharmacomechanical coupling of smooth muscle, which second messenger typically increases?

- (A) Cyclic adenosine monophosphate
- (B) Diacylglycerol lipid
- (C) Nitric oxide radical
- (D) Inositol trisphosphate
- (E) Cyclic guanosine monophosphate

- 24.** Which feature is common to both skeletal muscle and smooth muscle?
- (A) Calcium rise is needed for contraction
 - (B) Action potential arises spontaneously
 - (C) Troponin C regulatory subunit is present
 - (D) Actin thin filament is entirely absent
 - (E) Caveolae are present on the membrane
- 25.** A bronchodilator drug used in asthma most likely works by increasing which second messenger?
- (A) Intracellular calcium
 - (B) Inositol trisphosphate
 - (C) Diacylglycerol lipid
 - (D) Cyclic adenosine monophosphate
 - (E) Cyclic guanosine monophosphate
- 26.** In smooth muscle contraction, which event precedes the binding of calcium to calmodulin?
- (A) Calcium is pumped back by the calcium pump
 - (B) Myosin light chain kinase phosphorylates myosin
 - (C) Myosin and actin cross bridges are formed
 - (D) Calcium is released from the internal store
 - (E) Myosin phosphatase enzyme is activated
- 27.** Type I muscle fibres perform better than Type IIb fibres in marathon running due to their higher what?
- (A) Fibre diameter
 - (B) Peak force output
 - (C) Oxidative capacity
 - (D) Glycolytic capacity
 - (E) Contraction speed
- 28.** When someone lifts weights regularly, their muscles increase in size primarily through which mechanism?
- (A) More myofibrils are added inside existing fibres
 - (B) Adjacent fibres fuse into new and larger fibres
 - (C) The number of bony attachment sites rises
 - (D) Only the number of satellite cells increases
 - (E) The number of motor neurons supplying rises
- 29.** Which statement comparing different muscle types is true?
- (A) Smooth muscle lacks the troponin protein
 - (B) Skeletal muscle has pacemaker type cells
 - (C) Smooth muscle uses mechanical coupling
 - (D) Cardiac muscle lacks any gap junctions
 - (E) Cardiac muscle has no transverse tubule
- 30.** In tetanus toxin action, which mechanism is impaired, leading to muscle spasms?
- (A) Dihydropyridine receptor opening is lost
 - (B) Acetylcholine synthesis is blocked at end
 - (C) Myosin light chain kinase is inhibited
 - (D) Sarcoplasmic calcium pump is inhibited
 - (E) Inhibitory transmitter release is blocked
- 31.** The muscle fibres of a bodybuilder versus a marathon runner primarily differ in which major trait?
- (A) Only in the density of capillaries
 - (B) They have same myoglobin content
 - (C) They have same enzyme activity
 - (D) They have same mitochondria count
 - (E) Glycolytic against oxidative capacity

- 32.** During a thyroidectomy, the superior thyroid artery is ligated close to the gland to avoid injuring which nerve?
- (A) External branch of superior laryngeal nerve
 - (B) Phrenic nerve on the anterior scalene muscle
 - (C) Internal branch of superior laryngeal nerve
 - (D) Main trunk of vagus nerve within carotid sheath
 - (E) Recurrent laryngeal branch of vagus nerve
- 33.** Which cranial nerve pair supplies motor fibres to the posterior belly of the digastric and the stylohyoid muscles?
- (A) Facial nerve and hypoglossal nerve
 - (B) Facial nerve and glossopharyngeal nerve
 - (C) Trigeminal nerve and facial nerve
 - (D) Facial nerve and vagus nerve
 - (E) Glossopharyngeal nerve and vagus nerve
- 34.** For an internal jugular venipuncture, which landmark is commonly used?
- (A) One inch anterior to the mandible angle
 - (B) One inch above the manubrium of sternum
 - (C) Midpoint of posterior border of the muscle
 - (D) A point lateral to the thyroid cartilage
 - (E) Triangle between the two heads of muscle
- 35.** Regarding a thyroidectomy, which statement is incorrect?
- (A) The gland is covered by the middle cervical fascia
 - (B) The gland lies below the thyroid cartilage level
 - (C) Hormone replacement is always needed after this
 - (D) The recurrent laryngeal nerve is at risk of injury
 - (E) Supply is from superior and inferior thyroid artery
- 36.** Which cranial nerve pairing with its function in the neck is incorrect?
- (A) Vagus nerve supplies the larynx and its sensation
 - (B) Accessory nerve gives motor supply to trapezius muscle
 - (C) Hypoglossal nerve supplies tongue except palatoglossus
 - (D) Glossopharyngeal nerve carries afferent from carotid sinus
 - (E) Facial nerve gives motor supply to the mylohyoid muscle
- 37.** During removal of a mole near the clavicle, which cutaneous nerve is at risk of injury?
- (A) Transverse cervical nerve
 - (B) Cervical branch of facial
 - (C) Great auricular nerve
 - (D) Lesser occipital nerve
 - (E) Supraclavicular nerve
- 38.** For a cardiac catheterisation procedure, which blood vessel is most commonly accessed?
- (A) External jugular vein
 - (B) Subclavian artery
 - (C) Common carotid artery
 - (D) Internal jugular vein
 - (E) Femoral artery
- 39.** Which of these muscles does not help to elevate the larynx?
- (A) Mylohyoid muscle
 - (B) Sternothyroid muscle
 - (C) Thyrohyoid muscle
 - (D) Stylohyoid muscle
 - (E) Digastric muscle
- 40.** An interscalene block primarily affects which structure?
- (A) Brachial plexus
 - (B) Accessory nerve
 - (C) Cervical plexus
 - (D) Phrenic nerve
 - (E) Sympathetic trunk

- 41.** At which of these sites would you not typically be able to feel a pulse?
- (A) Inferior thyroid artery
 - (B) Facial artery at mandible
 - (C) Superficial temporal artery
 - (D) Radial artery at the wrist
 - (E) Common carotid artery
- 42.** Which surface line helps to localise the external jugular vein?
- (A) Between the two heads of sternocleidomastoid
 - (B) From the angle of mandible to the mid clavicle
 - (C) From the jugular notch up to the ear lobule
 - (D) From the midline of neck to mastoid process
 - (E) Along the posterior border of the same muscle
- 43.** Which large blood vessel is commonly used for central venous catheterisation?
- (A) External jugular vein
 - (B) Brachial vein
 - (C) Internal jugular vein
 - (D) Femoral artery
 - (E) Radial artery
- 44.** Pain and swelling at the tip of the chin would suggest enlargement of which lymph nodes?
- (A) Deep cervical nodes
 - (B) Submental nodes
 - (C) Submandibular nodes
 - (D) Parotid nodes
 - (E) Preauricular nodes

03 SCALP, FACE, PAROTID REGION, EAR, CRANIAL FOSSA

LECTURES 3-4

- 45.** A tumour eroding the floor of the middle ear is most likely to affect which structure?
- (A) Sigmoid venous sinus
 - (B) Internal carotid artery
 - (C) Aditus to mastoid antrum
 - (D) Internal jugular vein
 - (E) Cochlea of inner ear
- 46.** In a middle cranial fossa fracture, which nerve would not likely be injured?
- (A) Abducens nerve
 - (B) Oculomotor nerve
 - (C) Trochlear nerve
 - (D) Hypoglossal nerve
 - (E) Trigeminal nerve
- 47.** In cavernous sinus thrombosis, which nerve is not typically affected?
- (A) Trochlear nerve of the orbit
 - (B) Mandibular division of trigeminal
 - (C) Maxillary division of trigeminal
 - (D) Ophthalmic division of trigeminal
 - (E) Oculomotor nerve of the orbit
- 48.** In cavernous sinus thrombosis, which cranial nerve is characteristically affected first?
- (A) Optic nerve
 - (B) Mandibular nerve
 - (C) Olfactory nerve
 - (D) Facial nerve
 - (E) Abducens nerve

- 49.** If the mandibular branch of the facial nerve is cut, which muscle would be paralysed?
- (A) Orbicularis oris muscle
 - (B) Levator labii superioris
 - (C) Zygomaticus major muscle
 - (D) Buccinator muscle
 - (E) Depressor anguli oris
- 50.** If a patient experiences saliva pooling in their mouth and drooling, it suggests paralysis of which muscle?
- (A) Buccinator muscle
 - (B) Masseter muscle
 - (C) Orbicularis oris
 - (D) Zygomaticus major
 - (E) Levator anguli oris
- 51.** Which layer of the scalp is considered the danger area because infection spreads easily through it?
- (A) Aponeurotic layer
 - (B) Pericranium layer
 - (C) Loose areolar tissue
 - (D) Dense connective tissue
 - (E) Skin layer of scalp
- 52.** Removing ear wax can sometimes trigger a cough. Which nerve is responsible for this reflex?
- (A) Facial nerve
 - (B) Glossopharyngeal nerve
 - (C) Vestibulocochlear nerve
 - (D) Vagus nerve
 - (E) Trigeminal nerve
- 53.** Where is a myringotomy best performed?
- (A) At the central umbo point
 - (B) Posterior superior quadrant
 - (C) Anterior superior quadrant
 - (D) Anterior inferior quadrant
 - (E) Posterior inferior quadrant
- 54.** How does an infection in the nasopharynx spread to the middle ear?
- (A) Through the mastoid air cell system
 - (B) Through the pharyngotympanic tube
 - (C) Through the external acoustic meatus
 - (D) Through the semicircular canal system
 - (E) Through the internal acoustic meatus
- 55.** An epidural haematoma most often occurs after the rupture of which vessel?
- (A) Superior sagittal sinus
 - (B) Middle meningeal artery
 - (C) Basilar artery of brain
 - (D) Anterior cerebral artery
 - (E) Bridging cerebral vein
- 56.** In a severe infection of the cavernous sinus, which cranial nerve is most likely to be spared?
- (A) Ophthalmic division of trigeminal
 - (B) Oculomotor nerve of the orbit
 - (C) Abducens nerve of the orbit
 - (D) Mandibular division of trigeminal
 - (E) Trochlear nerve of the orbit
- 57.** Procedures on the tympanic membrane carry a risk of injuring which nearby nerve?
- (A) Lesser petrosal nerve
 - (B) Glossopharyngeal nerve
 - (C) Facial nerve main trunk
 - (D) Chorda tympani nerve
 - (E) Vagus nerve trunk

58. Children are more prone to middle ear infection partly because their pharyngotympanic tube is which of the following?

- (A) Wider and more vertical
- (B) Narrower and more vertical
- (C) Longer and more vertical
- (D) Obliterated fully at birth
- (E) Shorter and more horizontal

04 DEVELOPMENT OF HEAD AND NECK

LECTURE 5

59. Which cranial nerve innervates derivatives of the first pharyngeal arch?

- (A) Hypoglossal nerve
- (B) Trigeminal nerve
- (C) Glossopharyngeal nerve
- (D) Facial nerve
- (E) Vagus nerve

60. Which cranial nerve innervates derivatives of the second pharyngeal arch?

- (A) Trigeminal nerve
- (B) Hypoglossal nerve
- (C) Facial nerve
- (D) Glossopharyngeal nerve
- (E) Vagus nerve

61. Which of the following is not a primary component of a pharyngeal arch?

- (A) A muscular component
- (B) An aortic arch artery
- (C) A cartilaginous bar rod
- (D) A lymph node of neck
- (E) A specific cranial nerve

62. A defect from failure of the intermaxillary process to form would likely spare which structure?

- (A) Forehead of the face
- (B) Premaxilla of upper jaw
- (C) Nasal septum of the nose
- (D) Philtrum of the upper lip
- (E) Primary palate of mouth

63. A unilateral cleft lip results from the failed fusion of which structures?

- (A) Lateral palatal process with the primary palate
- (B) Lateral palatal processes with one another
- (C) Medial nasal prominence with maxillary one
- (D) Medial nasal prominence with lateral nasal one
- (E) Lateral nasal prominence with maxillary one

64. A bridge of tissue that can sometimes be found across a cleft lip is called what?

- (A) Simonart band
- (B) Bilateral cleft lip
- (C) Anterior cleft palate
- (D) Posterior cleft palate
- (E) Complete cleft palate

65. The persistence of the thyroglossal duct can produce which condition?

- (A) Thyroglossal duct cyst
- (B) Branchial fistula of neck
- (C) Pierre Robin sequence
- (D) Ankyloglossia of tongue
- (E) Treacher Collins syndrome

- 66.** The failure of the cervical sinus to obliterate leads to which condition?
- (A) Holoprosencephaly of brain
 - (B) Branchial fistula of neck
 - (C) Pierre Robin sequence
 - (D) Cervical thyroglossal cyst
 - (E) Lingual cyst of tongue
- 67.** A cleft tongue is due to the failed fusion of which structures?
- (A) Epiglottal swelling
 - (B) Copula of the tongue
 - (C) Hypopharyngeal eminence
 - (D) Maxillary prominences
 - (E) Lateral lingual swellings
- 68.** What is the term for a short lingual frenulum which can restrict tongue movement?
- (A) Cleft tongue
 - (B) Glossoptosis
 - (C) Ankyloglossia
 - (D) Microglossia
 - (E) Macroglossia
- 69.** The first pharyngeal groove develops into which structure?
- (A) Middle ear cavity space
 - (B) Tympanic membrane layer
 - (C) Mastoid air cell system
 - (D) Pharyngotympanic tube
 - (E) External acoustic meatus
- 70.** Treacher Collins syndrome primarily affects the development of which structure?
- (A) Greater horn of hyoid
 - (B) Stapes of middle ear
 - (C) Hyoid bone of neck
 - (D) Laryngeal cartilages
 - (E) Mandible of the face
- 71.** The posterior one third of the tongue develops from which embryonic primordium?
- (A) Tuberculum impar bud
 - (B) Lateral lingual swellings
 - (C) Hypopharyngeal eminence
 - (D) Epiglottic swelling
 - (E) Median tongue bud

05 TEMPORAL AND INFRATEMPORAL FOSSA, TEMPOROMANDIBULAR JOINT

LECTURE 6

- 72.** Which of the following is not a branch of the second part of the maxillary artery?
- (A) Buccal artery
 - (B) Inferior alveolar artery
 - (C) Pterygoid artery
 - (D) Masseteric artery
 - (E) Deep temporal artery
- 73.** If someone fractures their zygomatic arch, which muscle is most likely to be paralysed?
- (A) Buccinator muscle
 - (B) Medial pterygoid
 - (C) Masseter muscle
 - (D) Lateral pterygoid
 - (E) Temporalis muscle

- 74.** A patient with tetanus cannot open their mouth. Which muscle is primarily affected?
- (A) Medial pterygoid
 - (B) Lateral pterygoid
 - (C) Temporalis muscle
 - (D) Buccinator muscle
 - (E) Masseter muscle
- 75.** Which of these is a sensory branch of the mandibular nerve?
- (A) Lateral pterygoid nerve
 - (B) Medial pterygoid nerve
 - (C) Masseteric nerve
 - (D) Deep temporal nerve
 - (E) Buccal nerve
- 76.** Which artery passes through the foramen ovale to reach the trigeminal ganglion?
- (A) Inferior alveolar artery
 - (B) Ascending pharyngeal artery
 - (C) Accessory meningeal artery
 - (D) Middle meningeal artery
 - (E) Deep auricular artery
- 77.** If a patient reports numbness from under their eye down to below their nose, where is the lesion likely located?
- (A) Apex of the orbit
 - (B) Cavernous sinus
 - (C) Temporal fossa
 - (D) Pterygopalatine fossa
 - (E) Infratemporal fossa
- 78.** A fracture of the coronoid process of the mandible would injure which muscle?
- (A) Lateral pterygoid
 - (B) Buccinator muscle
 - (C) Masseter muscle
 - (D) Medial pterygoid
 - (E) Temporalis muscle
- 79.** Which of these is not a branch of the anterior division of the mandibular nerve?
- (A) Lateral pterygoid nerve
 - (B) Masseteric nerve
 - (C) Medial pterygoid nerve
 - (D) Deep temporal nerve
 - (E) Buccal nerve
- 80.** To move the jaw to the right side, which muscles are required?
- (A) Left masseter and left temporalis
 - (B) Both lateral pterygoid together
 - (C) Left lateral and left medial pterygoid
 - (D) Right lateral and right medial pterygoid
 - (E) Right masseter and right temporalis
- 81.** Which of these is not considered a boundary of the infratemporal fossa?
- (A) Medial pterygoid plate
 - (B) Lateral pterygoid plate
 - (C) Zygomatic bone of face
 - (D) Mastoid and styloid process
 - (E) Ramus of the mandible
- 82.** Which artery is a branch of the second part of the maxillary artery?
- (A) Deep auricular artery
 - (B) Pterygoid muscular branch
 - (C) Inferior alveolar artery
 - (D) Middle meningeal artery
 - (E) Infraorbital artery

83. Which artery enters the mandibular foramen to supply the lower teeth?

- (A) Posterior superior alveolar
- (B) Deep auricular artery
- (C) Accessory meningeal artery
- (D) Inferior alveolar artery
- (E) Middle meningeal artery

06 EXTRAOCULAR MUSCLES AND INNERVATION

LECTURE 7

84. A patient's eye is abducted and looks slightly downward at rest. This suggests a palsy of which nerve?

- (A) Abducens nerve of the orbit
- (B) Superior division of oculomotor
- (C) Inferior division of oculomotor
- (D) Trochlear nerve of the orbit
- (E) Optic nerve of the eyeball

85. Which nerve supplies the medial rectus muscle?

- (A) Abducens nerve of the orbit
- (B) Oculomotor nerve main trunk
- (C) Inferior division of oculomotor
- (D) Trochlear nerve of the orbit
- (E) Superior division of oculomotor

86. Which muscle rotates the eyeball laterally?

- (A) Medial rectus muscle
- (B) Lateral rectus muscle
- (C) Superior rectus muscle
- (D) Superior oblique muscle
- (E) Inferior rectus muscle

87. During an eye movement test, a patient is unable to abduct the left eye. Which nerve is likely damaged?

- (A) Left oculomotor nerve trunk
- (B) Left abducens nerve of orbit
- (C) Left superior division of third
- (D) Left trochlear nerve of orbit
- (E) Left inferior division of third

88. What actions does the superior rectus muscle primarily perform?

- (A) Elevates, adducts, and intorts
- (B) Elevates the eyeball only
- (C) Elevates, abducts, and extorts
- (D) Depresses, adducts, and intorts
- (E) Depresses, abducts, and extorts

89. What are the main actions of the inferior oblique muscle?

- (A) Elevates and adducts the eye
- (B) Elevates and abducts the eye
- (C) Depresses and abducts eye
- (D) Depresses and adducts eye
- (E) Adducts the eyeball only

90. A patient tries to look left, but their right eye cannot adduct. Which nerve is likely damaged?

- (A) Abducens nerve of the orbit
- (B) Superior division of oculomotor
- (C) Inferior division of oculomotor
- (D) Trochlear nerve of the orbit
- (E) Oculomotor nerve main trunk

91. Which structures pass through the optic canal?

- (A) Ophthalmic vein and artery
- (B) Optic vein and ophthalmic artery
- (C) Optic nerve and ophthalmic vein
- (D) Optic nerve and ophthalmic artery
- (E) Emissary vein and optic nerve

92. If a patient's eye is persistently positioned down and out, which nerve is likely affected?

- (A) Abducens nerve of the orbit
- (B) Inferior division of oculomotor
- (C) Superior division of oculomotor
- (D) Oculomotor nerve main trunk
- (E) Trochlear nerve of the orbit

93. What actions does the inferior rectus muscle primarily perform?

- (A) Depresses, adducts, and extorts
- (B) Elevates, adducts, and intorts
- (C) Depresses, abducts, and intorts
- (D) Elevates, abducts, and extorts
- (E) Depresses the eyeball only

07 THORACIC WALL AND MUSCLES

LECTURE 8

94. Which statement about the ribs and intercostal muscles is correct?

- (A) The intercostal muscles are supplied by internal thoracic nerve
- (B) The first rib is crossed by the vein in front of scalene tubercle
- (C) The internal intercostal fibres run in an antero inferior manner
- (D) The first rib joins the transverse process of the seventh cervical
- (E) The neurovascular bundle lies between external and internal layer

95. Which muscle does not assist in normal quiet inspiration?

- (A) Serratus posterior superior
- (B) Levator costarum muscle
- (C) External intercostal muscle
- (D) Diaphragm of the thorax
- (E) Transversus thoracis muscle

96. Which structure does not pass through the superior thoracic aperture?

- (A) Left common carotid artery
- (B) Larynx of the airway
- (C) Recurrent laryngeal nerve
- (D) Thoracic duct of the body
- (E) Phrenic nerve of the neck

97. How many pairs of ribs articulate directly with the sternum?

- (A) Three pairs of ribs
- (B) Five pairs of ribs
- (C) Seven pairs of ribs
- (D) Eleven pairs of ribs
- (E) Nine pairs of ribs

98. The tubercle of the fourth rib articulates with which structure?

- (A) Body of third thoracic vertebra
- (B) Transverse process of the fifth
- (C) Transverse process of the fourth
- (D) Body of second thoracic vertebra
- (E) Body of first thoracic vertebra

99. The first and second posterior intercostal arteries arise from which artery?

- (A) Superior intercostal artery
- (B) Thoracic aorta of chest
- (C) Internal thoracic artery
- (D) Musculophrenic artery
- (E) Superior epigastric artery

100. The sixth to tenth posterior intercostal arteries arise from which vessel?

- (A) Musculophrenic artery
- (B) Thoracic aorta of chest
- (C) Internal thoracic artery
- (D) Superior epigastric artery
- (E) Superior intercostal artery

101. The tubercle of a typical rib articulates with which part of a vertebra?

- (A) The transverse process of third and fourth
- (B) The body of the fifth thoracic vertebra
- (C) The bodies of second and third thoracic
- (D) The transverse process of the same level
- (E) The body of the fourth thoracic vertebra

102. A stab wound two centimetres deep at the right fifth costal cartilage, three centimetres lateral to the sternum, would injure which artery?

- (A) Internal thoracic and intercostal
- (B) Anterior intercostal and musculophrenic
- (C) Internal thoracic and musculophrenic
- (D) Thoracic aorta of the chest wall
- (E) Musculophrenic and subcostal artery

103. Which of these muscles does not elevate the ribs during breathing?

- (A) Internal intercostal muscle
- (B) Levator costarum muscle
- (C) External intercostal muscle
- (D) Transversus thoracis muscle
- (E) Serratus posterior superior

104. Which structure is not located at the level of the sternal angle?

- (A) The entry of the azygos vein into vein
- (B) The disc between fourth and fifth
- (C) The beginning of the ascending aorta
- (D) The highest point of the aortic arch
- (E) The bifurcation point of the trachea

105. During a dissection, at the lateral border of the sternum, which muscle would you see?

- (A) Internal intercostal muscle
- (B) Innermost intercostal muscle
- (C) Levator costarum muscle
- (D) Transversus thoracis muscle
- (E) External intercostal muscle

106. The body of the sternum typically corresponds to which vertebral levels?

- (A) Fourth to sixth thoracic
- (B) Fifth to ninth thoracic
- (C) Ninth to twelfth thoracic
- (D) Second to seventh thoracic
- (E) First to fourth thoracic

107. An injury at the thoracic inlet would endanger which artery?

- (A) Brachial artery
- (B) Axillary artery
- (C) Internal carotid artery
- (D) Subclavian artery
- (E) Common carotid artery

108. A bullet penetrating the eighth intercostal space would likely injure which artery?

- (A) Musculophrenic artery
- (B) Posterior intercostal artery
- (C) Superior epigastric artery
- (D) Internal thoracic artery
- (E) Anterior intercostal artery

109. Which muscle attaches to the fifth rib near the sternum but lies posteriorly?

- (A) Levator costarum muscle
- (B) Transversus thoracis muscle
- (C) Innermost intercostal muscle
- (D) Serratus anterior muscle
- (E) External intercostal muscle

110. Which movement is produced mainly by the serratus anterior muscle?

- (A) Retraction of the scapula
- (B) Upward rotation of scapula
- (C) Downward rotation of bone
- (D) Depression of the scapula
- (E) Elevation of the scapula

08 ANTERIOR ABDOMINAL WALL AND HERNIA

LECTURE 9

111. Concerning the inguinal region, which statement is incorrect?

- (A) The superficial ring passes the round ligament
- (B) The transversalis fascia is the posterior wall
- (C) The lacunar ligament is medial to femoral ring
- (D) The inguinal ligament comes from external oblique
- (E) The conjoint tendon comes from two muscle layers

112. A failure of the processus vaginalis to close after birth leads to which condition?

- (A) Femoral hernia of thigh
- (B) Indirect inguinal hernia
- (C) Direct inguinal hernia
- (D) Epigastric hernia of wall
- (E) Umbilical hernia of navel

113. A subcostal incision for a cholecystectomy would most likely endanger which spinal nerves?

- (A) Eighth and ninth thoracic
- (B) Seventh and eighth thoracic
- (C) Ninth and tenth thoracic
- (D) Eleventh and twelfth thoracic
- (E) Fifth and sixth thoracic

114. Which statement regarding the anatomy of the abdominal wall is correct?

- (A) At third lumbar level a layer is behind the muscle
- (B) Internal oblique aponeurosis forms inguinal ligament
- (C) The internal oblique muscle inserts on lower three ribs
- (D) Superficial and inferior epigastric join in the sheath
- (E) The pyramidalis muscle is supplied by subcostal nerve

115. A knife stab three centimetres deep two centimetres below and medial to the umbilicus would least likely injure which structure?

- (A) Rectus abdominis muscle
- (B) Posterior rectus sheath
- (C) Twelfth thoracic nerve
- (D) Inferior epigastric vessel
- (E) Anterior rectus sheath

116. A horizontal knife cut four centimetres deep through the umbilicus would least likely injure which structure?

- (A) Posterior rectus sheath
- (B) Anterior rectus sheath
- (C) Tenth thoracic nerve
- (D) Deep circumflex iliac vessel
- (E) Rectus abdominis muscle

117. Which statement about the inguinal region is incorrect?

- (A) External oblique makes the superficial ring
- (B) Testicular artery and duct are inside the cord
- (C) Genitofemoral nerve supplies cremaster muscle
- (D) Indirect hernia sac passes the whole canal
- (E) Direct hernia sac lies lateral to the artery

118. Which statement about the inguinal region is incorrect?

- (A) Superficial ring is in the external oblique layer
- (B) Genitofemoral nerve supplies cremaster and skin
- (C) Testicular artery and duct pass the whole canal
- (D) Scrotal lymph drains to superficial inguinal node
- (E) Internal spermatic fascia comes from the muscle

119. Which statement regarding an indirect inguinal hernia is correct?

- (A) It passes through the whole inguinal canal
- (B) It is due to weakness of inguinal triangle
- (C) It herniates through medial inguinal fossa
- (D) It is most commonly found in the elderly
- (E) It exits the ring lateral to spermatic cord

120. Which statement regarding a direct inguinal hernia is correct?

- (A) It exits the ring inside the spermatic cord
- (B) It passes through the whole inguinal canal
- (C) It is due to weakness of inguinal triangle
- (D) It is caused by patent processus vaginalis
- (E) It originates in the lateral inguinal fossa

121. Loss of the cremasteric reflex indicates damage to the motor fibres of which nerve?

- (A) Genitofemoral nerve
- (B) Pudendal nerve
- (C) Femoral nerve
- (D) Ilioinguinal nerve
- (E) Obturator nerve

122. Regarding the abdominal wall, which of these statements is incorrect?

- (A) At third lumbar level a layer lies behind muscle
- (B) External oblique makes the inguinal ligament
- (C) Cremaster muscle comes from internal oblique
- (D) Pyramidalis muscle is supplied by subcostal
- (E) Inferior epigastric artery comes from femoral

123. A suprapubic incision would least likely injure which structure?

- (A) Linea alba of the midline
- (B) Deep circumflex iliac artery
- (C) Arcuate line of the sheath
- (D) Pyramidalis muscle of wall
- (E) Inferior epigastric artery

124. Lymph fluid from a suprapubic drain would flow to which lymph nodes?

- (A) Superficial inguinal node
- (B) Para-aortic node
- (C) Deep inguinal node
- (D) Axillary node of armpit
- (E) External iliac node

- 125.** A groin hydrocele often seen with an indirect hernia is caused by the persistence of what?
- (A) Processus vaginalis
 - (B) Patent umbilical vein
 - (C) Femoral ring weakness
 - (D) Dilated pampiniform plexus
 - (E) Weak transversalis fascia
- 126.** Which statement about the floor of the inguinal canal is incorrect?
- (A) The deep ring is a gap in transversalis fascia
 - (B) The conjoint tendon strengthens the back wall
 - (C) The inguinal ligament itself forms the canal floor
 - (D) The external oblique aponeurosis makes the floor
 - (E) The transversalis fascia makes the posterior wall
- 127.** A skin infection at McBurney's point would drain its lymph to which nodes?
- (A) Superficial inguinal node
 - (B) Para-aortic node of back
 - (C) Deep inguinal node
 - (D) Axillary node of armpit
 - (E) External iliac node
- 128.** Which fascia layer of the spermatic cord is derived from the external oblique aponeurosis?
- (A) Transversalis fascia layer
 - (B) Dartos fascia of scrotum
 - (C) External spermatic fascia
 - (D) Internal spermatic fascia
 - (E) Cremasteric fascia layer
- 129.** Where does a femoral hernia typically emerge?
- (A) Medial to pubic tubercle and above ligament
 - (B) Lateral to pubic tubercle and below ligament
 - (C) Lateral to pubic tubercle and above ligament
 - (D) Through the superficial inguinal ring only
 - (E) Medial to pubic tubercle and below ligament
- 130.** Which statement correctly contrasts a direct versus an indirect inguinal hernia?
- (A) Direct hernia traverses the whole inguinal canal
 - (B) Indirect hernia exits via Hesselbach triangle
 - (C) Direct hernia enters the deep inguinal ring
 - (D) Indirect hernia exits medial to pubic tubercle
 - (E) Direct hernia passes medial to epigastric vessel
- 131.** During a Caesarean section, which structure is usually spared from injury with a common lower transverse incision?
- (A) Round ligament of uterus
 - (B) Ilioinguinal nerve of wall
 - (C) Iliohypogastric nerve
 - (D) Inferior epigastric vessels
 - (E) Uterine artery of pelvis

Answer key & reasoning

Why the right answer is right — and where the past paper is wrong

Each answer states the reasoning, not just the fact. Where the original past paper contains an **outright error**, it is marked in red and corrected — do not memorise the old key on those items.

01 MUSCLE PHYSIOLOGY

LECTURES 10-12

C Increasing the opening time of the ryanodine receptor

Longer ryanodine receptor opening means more calcium released from the sarcoplasmic reticulum, so more calcium reaches troponin C and force rises. The dihydropyridine channel is the trigger but the bulk of calcium comes from the store. Faster calcium pump activity and greater sodium calcium exchange both remove calcium, which lowers force.

C Troponin

Smooth muscle has no troponin at all. Its thin filament regulation works through calcium binding calmodulin, which activates myosin light chain kinase (thick filament regulation). Tropomyosin, dense bodies, calmodulin, and gap junctions are all present.

E Membrane depolarisation of the cell

Membrane depolarisation opens voltage gated calcium channels, calcium enters and triggers further release, then binds calmodulin, activating the kinase, which phosphorylates the myosin light chain, letting myosin attach to actin. Depolarisation is the first event in the electromechanical chain.

A Calcium induced calcium release

is the mechanism in cardiac and smooth muscle, but skeletal muscle uses direct mechanical coupling between the dihydropyridine receptor and the ryanodine receptor. All three have actin, myosin, and need adenosine triphosphate.

A At the junction of A and I bands

In mammalian skeletal muscle the transverse tubule sits at the junction of the A and I bands, forming a triad with two terminal cisternae. (In cardiac muscle the tubule lies at the Z line.)

A The sarcolemma costamere

Dystrophin links the actin cytoskeleton to the dystroglycan complex at the costamere on the sarcolemma. Without it, the membrane tears during contraction, calcium floods in, and the fibre degenerates. This is the basis of Duchenne muscular dystrophy.

B Sarcoplasmic reticulum

Even though smooth muscle relies heavily on extracellular calcium entry, its intracellular store is still the sarcoplasmic reticulum. Terminal cisternae are a skeletal muscle feature; caveolae are membrane invaginations, not storage sites.

C Degradation exceeding synthesis

Atrophy is a shift in protein balance: ubiquitin proteasome degradation outpaces synthesis. There is no loss of fibres or neurons; the existing fibres simply shrink.

D The sarcolemma is the plasma cell membrane

The sarcolemma is the muscle cell plasma membrane. A triad is one tubule plus two cisternae (not the reverse). Caveolae belong to smooth muscle. Calcium is stored in the sarcoplasmic reticulum, not the sarcomere.

B Phospholamban protein

Phospholamban inhibits the calcium pump when unphosphorylated. Beta adrenergic stimulation phosphorylates it, releasing the brake, so calcium is taken up faster (giving both faster relaxation and a larger store for the next beat).

A The cells are each innervated individually

Multi-unit smooth muscle (iris, vas deferens, large airways, piloerector) has few or no gap junctions; each cell receives its own nerve ending, allowing fine graded control. Gap junction coupling and cell to cell spread describe single unit (visceral) smooth muscle.

D Sodium channels of membrane open

Order: nerve action potential → acetylcholine release → endplate potential → sodium channels open on the sarcolemma → action potential spreads into the tubule → tubule depolarises → calcium release channels open → calcium binds troponin C → cross bridges form → calcium is pumped back. Only sodium channel opening comes before tubule depolarisation.

D A lack of adenosine triphosphate

Detaching the myosin head from actin requires adenosine triphosphate binding. After death, production ceases, heads stay locked onto actin, and the muscle stiffens.

E Calcium induced calcium release

Cardiac muscle depends on calcium induced calcium release; skeletal muscle uses direct mechanical coupling. Both have actin, troponin, and require a calcium rise triggered by an action potential.

D Myosin light chain kinase is activated

The calcium calmodulin complex directly binds and activates myosin light chain kinase. Phosphorylation of the myosin light chain then follows.

A By decreasing the phospholamban activity

Decreasing phospholamban activity removes the inhibition on the calcium pump, so the store loads more calcium and releases more per beat. Blocking the ryanodine receptor, reducing calcium induced release, more sodium calcium exchange, and shorter dihydropyridine opening all reduce force.

E The Z line anchors the thin filament ends

The Z line (Z disc) is the anchoring point for thin filaments; the M line anchors thick filaments. The A band is the full thick filament length, the H band is thick filament only (no overlap), and it is myosin, not actin, that hydrolyses adenosine triphosphate.

A Myosin adenosine triphosphatase activity

Fibre typing is classically based on the myosin adenosine triphosphatase isoform, which sets shortening velocity. Diameter, capillary density, myoglobin, and glycogen are correlated consequences, not the defining criterion.

B Muscle mass is lost slowly

Disuse causes atrophy: the fibres shrink through protein degradation. Synthesis falls rather than rises, cell number does not change (no hyperplasia), and disuse tends to shift fibres toward the fast type, not toward type one.

A Squeezing buttock without moving

Isometric means tension develops with no change in muscle length and no joint movement. Squeezing the buttock without moving the hip fits exactly. Walking uphill and rising from a chair are concentric; lowering a dumbbell is eccentric.

A Small units resistant to fatigue

The size principle: small motor neurons with the lowest threshold fire first, and these supply small, slow, fatigue resistant type one units. Large fast units are recruited last, only when high force is needed.

E Inhibiting the sodium potassium pump

Blocking the sodium potassium pump raises intracellular sodium, which reduces the gradient driving the sodium calcium exchanger. Less calcium is extruded, so more calcium is stored and released, raising force.

D Inositol trisphosphate

Pharmacomechanical coupling works through a G protein coupled receptor activating phospholipase C, which generates inositol trisphosphate. That releases calcium from the store without any change in membrane potential. Cyclic adenosine monophosphate, cyclic guanosine monophosphate, and nitric oxide cause relaxation.

A Calcium rise is needed for contraction

All muscle types require a rise in intracellular calcium to initiate contraction. Troponin C and spontaneous action potentials are not skeletal features shared with smooth muscle, caveolae are smooth muscle only, and both types have actin.

D Cyclic adenosine monophosphate

Beta two agonists stimulate adenyl cyclase, raising cyclic adenosine monophosphate in airway smooth muscle. This activates protein kinase A, which inhibits myosin light chain kinase and lowers calcium, giving relaxation and bronchodilation.

D Calcium is released from the internal store

Calcium must first be available in the cytosol before it can bind calmodulin. That calcium comes from entry across the membrane and release from the store. Kinase phosphorylation, cross bridge formation, and calcium reuptake all come after.

C Oxidative capacity

Type one fibres have high mitochondrial density, high myoglobin, and rich capillary supply, giving high oxidative capacity and fatigue resistance — ideal for sustained endurance. Type two b fibres have high peak force and speed but fatigue quickly.

A More myofibrils are added inside existing fibres

Hypertrophy: existing fibres add contractile protein, building more myofibrils and sarcomeres in parallel, so the fibre grows in cross sectional area. Human muscle does not meaningfully add new fibres.

A Smooth muscle lacks the troponin protein

Smooth muscle lacks troponin entirely and regulates contraction through calmodulin and myosin light chain kinase. Cardiac muscle does have transverse tubules and abundant gap junctions; skeletal muscle has no pacemaker cells; direct mechanical coupling is a skeletal (not smooth) feature.

E Inhibitory transmitter release is blocked

Tetanus toxin cleaves synaptobrevin in inhibitory interneurons of the spinal cord, blocking glycine and gamma aminobutyric acid release. Motor neurons lose inhibition and fire continuously, causing spasm and lockjaw. (Contrast botulinum toxin, which blocks acetylcholine release, causing flaccid paralysis.)

E Glycolytic against oxidative capacity

A bodybuilder's hypertrophied type two fibres favour anaerobic glycolysis for short bursts of high force; a marathon runner's type one fibres favour oxidative metabolism for endurance. They differ in myoglobin, mitochondria, capillaries, and enzyme activity as well.

A External branch of superior laryngeal nerve

The external branch of the superior laryngeal nerve runs alongside the superior thyroid artery near the upper pole and supplies the cricothyroid muscle. Ligating close to the gland separates the two. Injury gives a weak, low pitched voice. (The recurrent laryngeal nerve is at risk from the inferior thyroid artery at the lower pole.)

D Facial nerve and vagus nerve

Both the posterior belly of the digastric and the stylohyoid are second pharyngeal arch derivatives, so both are supplied by the facial nerve. (The anterior belly of the digastric is a first arch derivative supplied by the mandibular nerve.) Note: the paired option here is the one that includes the facial nerve for both; among the choices, only the facial nerve is correct for both muscles.

E Triangle between the two heads of muscle

The internal jugular vein lies deep to the triangle formed between the sternal and clavicular heads of the sternocleidomastoid, with the carotid artery medial to it. This is the standard central approach landmark.

C Hormone replacement is always needed after this

Hormone replacement is not always required — it depends on how much gland remains. After a hemithyroidectomy many patients stay euthyroid. All the other statements are true: the middle layer of deep cervical fascia (pretracheal) invests it, the recurrent laryngeal nerve is at risk, and the arterial supply is as described.

E Facial nerve gives motor supply to the mylohyoid muscle

The facial nerve supplies the posterior belly of the digastric and the stylohyoid. The mylohyoid and the anterior belly of the digastric are supplied by the nerve to mylohyoid from the mandibular division of the trigeminal nerve. All other pairings are correct.

E Supraclavicular nerve

The supraclavicular nerves (from the cervical plexus, third and fourth cervical roots) emerge at the posterior border of the sternocleidomastoid and cross the clavicle to supply the skin over it. The great auricular and lesser occipital nerves run upward instead.

E Femoral artery

The femoral artery is the classic access site: large, superficial, easily compressed against the femoral head after the procedure, and it gives a straight route up the aorta to the coronary ostia. (The radial artery is the other common site.)

B Sternothyroid muscle

The sternothyroid is an infrahyoid (strap) muscle that depresses the larynx. The stylohyoid, mylohyoid, and digastric are suprahyoid muscles that elevate it, and the thyrohyoid pulls the larynx up toward the hyoid.

A Brachial plexus

The interscalene block targets the brachial plexus roots and trunks in the groove between the anterior and middle scalene muscles, for shoulder and upper arm surgery. The phrenic nerve is frequently blocked as a side effect, but it is not the target.

A Inferior thyroid artery

The inferior thyroid artery is a deep branch of the thyrocervical trunk, lying behind the carotid sheath and gland — it is not palpable. The facial, superficial temporal, common carotid, and radial arteries are all standard pulse points.

B From the angle of mandible to the mid clavicle

The external jugular vein is drawn from the angle of the mandible, running obliquely down and back across the sternocleidomastoid, toward the middle of the clavicle, where it pierces the deep fascia to join the subclavian vein.

C Internal jugular vein

The internal jugular vein is the standard choice for central venous access: large calibre, predictable landmarks, straight path to the superior vena cava, and easily compressible. (The subclavian vein is the other common site.)

B Submental nodes

The submental nodes lie below the chin between the anterior bellies of the digastric muscles. They drain the tip of the tongue, the central lower lip, the floor of the mouth, and the chin.

03 SCALP, FACE, PAROTID REGION, EAR, CRANIAL FOSSA

LECTURES 3–4

D Internal jugular vein

The floor (jugular wall) of the middle ear cavity is a thin plate of bone separating it from the superior bulb of the internal jugular vein. Erosion there reaches the vein. (The carotid artery is the anterior wall, the aditus is the posterior wall, and the cochlea is the medial wall.)

D Hypoglossal nerve

The hypoglossal nerve exits through the hypoglossal canal in the posterior cranial fossa. The oculomotor, trochlear, ophthalmic, and abducens nerves run in the cavernous sinus wall and exit via the superior orbital fissure; the maxillary and mandibular nerves leave through the foramen rotundum and ovale — all middle fossa.

B Mandibular division of trigeminal

The mandibular division leaves the skull through the foramen ovale and never enters the cavernous sinus. The oculomotor, trochlear, ophthalmic, and maxillary nerves all lie in or on the lateral wall of the sinus (as does the abducens, within the sinus itself).

E Abducens nerve

The abducens nerve is the only one that runs free within the sinus cavity, directly beside the internal carotid artery, rather than embedded in the tougher lateral wall. That exposed course makes it the first to be compressed, so lateral gaze palsy is the earliest sign.

E Depressor anguli oris

The mandibular branch of the facial nerve supplies the depressor anguli oris, depressor labii inferioris, and mentalis. The buccinator and orbicularis oris are supplied by the buccal branch, and the zygomaticus major and levator labii superioris by the zygomatic and buccal branches.

A Buccinator muscle

The buccinator holds the cheek against the teeth and keeps food and saliva within the oral cavity. When paralysed, the cheek balloons out, food pools in the vestibule, and saliva dribbles out.

C Loose areolar tissue

Layer four, the loose areolar connective tissue, contains no strong fibrous barriers and carries the emissary veins that connect to the intracranial dural sinuses. Pus or blood can spread widely across the whole scalp and can carry infection into the cranium.

D Vagus nerve

The auricular branch of the vagus nerve (Arnold's nerve) supplies part of the external acoustic meatus. Stimulating it triggers the cough reflex through the vagal centre — sometimes called the ear cough reflex.

D Anterior inferior quadrant

The anterior inferior quadrant is the safest: it is away from the ossicular chain and the chorda tympani (postero superior), and it is the most dependent point for pus to drain. The postero superior quadrant is the danger zone.

B Through the pharyngotympanic tube

The pharyngotympanic (auditory) tube is the direct communication between the nasopharynx and the middle ear cavity. It normally equalises pressure, but it also carries organisms upward, which is why upper respiratory infection so often leads to middle ear infection.

B Middle meningeal artery

The middle meningeal artery runs in a groove on the inner table of the squamous temporal bone, at the pterion, which is the thinnest part of the skull. A blow there tears the artery and blood collects between bone and dura, giving the classic lucid interval.

D Mandibular division of trigeminal

The mandibular division exits the skull via the foramen ovale and does not enter the cavernous sinus, so it is spared. The abducens, oculomotor, trochlear, and ophthalmic nerves all pass through or along the sinus and are typically affected.

D Chorda tympani nerve

The chorda tympani, a branch of the facial nerve, crosses the medial surface of the tympanic membrane between the malleus and incus in the postero superior region. Injury causes loss of taste on the anterior two thirds of the tongue.

E Shorter and more horizontal

In children the tube is shorter, wider, and more horizontal, so organisms from the nasopharynx reach the middle ear easily and drainage is poor. It becomes longer and more oblique with growth of the skull base.

04 DEVELOPMENT OF HEAD AND NECK

LECTURE 5

B Trigeminal nerve

The trigeminal nerve is the nerve of the first (mandibular) arch, supplying the muscles of mastication, the mylohyoid, the anterior belly of the digastric, the tensor tympani, and the tensor veli palatini.

C Facial nerve

The facial nerve is the nerve of the second (hyoid) arch, supplying the muscles of facial expression, the stapedius, the stylohyoid, and the posterior belly of the digastric.

D A lymph node of neck

Every pharyngeal arch has four core components: a cartilage bar, a muscular component, a cranial nerve, and an aortic arch artery. Lymph nodes are not part of the arch plan.

A Forehead of the face

The intermaxillary segment forms the philtrum, the premaxilla with the four incisor teeth, the primary palate, and it contributes to the nasal septum. The forehead comes from the frontonasal prominence, so it is spared.

C Medial nasal prominence with maxillary one

Cleft lip occurs when the maxillary prominence fails to fuse with the medial nasal prominence on one side. This is the commonest orofacial cleft. Failure of the palatal shelves gives cleft palate instead.

A Simonart band

A Simonart band is a persisting bridge of soft tissue across the cleft, representing incomplete separation. It is seen in incomplete cleft lip.

A Thyroglossal duct cyst

The thyroglossal duct is the track along which the thyroid descends from the foramen caecum to the neck; it normally involutes. Persisting remnants form a midline cyst that moves upward on swallowing and on tongue protrusion.

B Branchial fistula of neck

The cervical sinus is the depression formed as the second arch overgrows arches three to six. Failure to obliterate leaves a branchial (lateral cervical) cyst or a fistula opening along the anterior border of the sternocleidomastoid.

E Lateral lingual swellings

The anterior two thirds of the tongue develops from two lateral lingual swellings plus the tuberculum impar. If the lateral swellings fail to merge, a midline groove or bifid tongue results.

C Ankyloglossia

(tongue tie) is a short, thick lingual frenulum that tethers the tongue tip to the floor of the mouth, restricting protrusion and sometimes interfering with feeding and speech.

E External acoustic meatus

The first pharyngeal groove (ectodermal cleft) deepens to become the external acoustic meatus. The tympanic membrane forms where it meets the first pouch, and the first pouch itself becomes the tube and middle ear cavity.

E Mandible of the face

Treacher Collins syndrome (mandibulofacial dysostosis) results from defective neural crest migration into the first arch, giving mandibular hypoplasia, malar hypoplasia, downslanting eyes, and ear defects.

C Hypopharyngeal eminence

The posterior third of the tongue develops from the hypopharyngeal eminence (copula), mainly third arch tissue, which is why its general sensation and taste are supplied by the glossopharyngeal nerve.

05 TEMPORAL AND INFRATEMPORAL FOSSA, TEMPOROMANDIBULAR JOINT

LECTURE 6

B Inferior alveolar artery

The inferior alveolar artery arises from the first (mandibular) part of the maxillary artery, along with the deep auricular, anterior tympanic, middle meningeal, and accessory meningeal arteries. The second (pterygoid) part gives the muscular branches: deep temporal, pterygoid, masseteric, and buccal.

C Masseter muscle

The masseter takes origin from the zygomatic arch. A fracture there disrupts the origin, so the masseter is functionally lost and jaw closure is weakened. (The temporalis passes deep to the arch and inserts on the coronoid process.)

E Masseter muscle

Trismus is spasm of the muscles of mastication that close the jaw: the masseter is the strongest and first affected, giving the classic lockjaw. The lateral pterygoid is the opener, not the closer.

E Buccal nerve

The buccal nerve is the only sensory branch of the anterior division of the mandibular nerve, carrying sensation from the cheek skin and mucosa. The masseteric, deep temporal, lateral pterygoid, and medial pterygoid nerves are all motor.

C Accessory meningeal artery

The accessory meningeal artery, a branch of the first part of the maxillary artery, ascends through the foramen ovale to supply the trigeminal ganglion and adjacent dura. (The middle meningeal artery passes through the foramen spinosum instead.)

D Pterygopalatine fossa

That skin territory — lower eyelid, cheek, side of nose, upper lip — is the maxillary division. It occupies the pterygopalatine fossa, passing from the foramen rotundum to the inferior orbital fissure, so a lesion there gives exactly this pattern.

E Temporalis muscle

The temporalis muscle inserts onto the coronoid process and the anterior border of the ramus. The masseter inserts on the lateral ramus and angle, the medial pterygoid on the medial angle, and the lateral pterygoid on the condylar neck.

C Medial pterygoid nerve

The nerve to the medial pterygoid arises from the main trunk of the mandibular nerve (not the anterior division), and it also supplies the tensor tympani and tensor veli palatini. The masseteric, deep temporal, lateral pterygoid, and buccal nerves all come from the anterior division.

C Left lateral and left medial pterygoid

Lateral excursion to the right is produced by the pterygoids of the opposite (left) side, which pull the left condyle forward and medially, swinging the chin to the right. This is why deviation on opening points toward the weak side.

C Zygomatic bone of face

The infratemporal fossa is bounded laterally by the ramus of the mandible, medially by the lateral pterygoid plate, anteriorly by the maxilla, posteriorly by the styloid and mastoid processes, and above by the greater wing of the sphenoid. The zygomatic bone is not a boundary. Note: the medial pterygoid plate is also not a true boundary in the strict sense, but the zygomatic bone is the intended answer as it is not part of any wall.

B Pterygoid muscular branch

The second (pterygoid) part of the maxillary artery gives only muscular branches: the pterygoid branches, deep temporal, masseteric, and buccal. The deep auricular, inferior alveolar, and middle meningeal come from the first part; the infraorbital from the third.

D Inferior alveolar artery

The inferior alveolar artery, from the first part of the maxillary artery, enters the mandibular foramen with the inferior alveolar nerve and runs in the mandibular canal to supply the lower teeth, emerging as the mental artery.

06 EXTRAOCULAR MUSCLES AND INNERVATION

LECTURE 7

C Inferior division of oculomotor

That division supplies the medial rectus, inferior rectus, and inferior oblique. Losing the medial rectus leaves the lateral rectus unopposed, so the eye drifts abducted; losing the inferior oblique (an elevator) leaves the superior oblique relatively unopposed, giving the slight downward tilt. A complete oculomotor palsy would instead give the full down and out eye with ptosis and a dilated pupil.

C Inferior division of oculomotor

The inferior division of the oculomotor nerve supplies the medial rectus, inferior rectus, and inferior oblique, plus the parasympathetic fibres to the ciliary ganglion. The superior division supplies only the superior rectus and levator palpebrae superioris.

B Lateral rectus muscle

The lateral rectus is the pure abductor, rotating the eyeball laterally, and it is supplied by the abducens nerve. The medial rectus is the pure adductor.

B Left abducens nerve of orbit

Abduction is performed only by the lateral rectus, which is supplied by the abducens nerve. Inability to abduct the left eye means the left abducens nerve is damaged, and the eye deviates medially at rest.

A Elevates, adducts, and intorts

Because the orbital axis lies about twenty three degrees lateral to the visual axis, the superior rectus pulls the eye up, in, and rotates it inward. Its primary action is elevation; the secondary actions are adduction and intorsion.

B Elevates and abducts the eye

The inferior oblique arises from the floor of the orbit anteriorly and runs backward and laterally to insert on the posterolateral globe. That line of pull elevates the eye, abducts it, and extorts it. Because it inserts laterally, it abducts (unlike the superior rectus, which adducts). It is the muscle tested by asking the patient to look up and out, and it is supplied by the inferior division of the oculomotor nerve.

Past paper error. The original key says the inferior oblique *adducts*. It does not. The muscle inserts on the posterolateral globe, so it **abducts**. Trust this key, not the old one.

C Inferior division of oculomotor

Adduction is performed by the medial rectus, supplied by the inferior division of the oculomotor nerve. An isolated failure of adduction with other functions intact points to that division.

D Optic nerve and ophthalmic artery

The optic canal, in the lesser wing of the sphenoid, transmits the optic nerve (with its meningeal sheaths) and the ophthalmic artery. The ophthalmic veins pass through the superior orbital fissure instead.

D Oculomotor nerve main trunk

A complete oculomotor palsy paralyses the superior, inferior, and medial recti and the inferior oblique, leaving only the lateral rectus (abducens) and superior oblique (trochlear) working. The eye therefore rests abducted and depressed, the classic down and out position, accompanied by ptosis from loss of the levator palpebrae superioris and a fixed dilated pupil from loss of the parasympathetic fibres.

A Depresses, adducts, and extorts

The inferior rectus, pulling along the orbital axis, depresses the eye and secondarily adducts and extorts it. Depression is its primary action.

07 THORACIC WALL AND MUSCLES

LECTURE 8

C The internal intercostal fibres run in an antero inferior manner

The internal intercostal fibres run downward and forward (antero inferior), at right angles to the external intercostals, which run downward and backward. The first rib joins the transverse process of the first thoracic (not the seventh cervical), the subclavian vein crosses in front of the scalene tubercle but the artery is behind, the neurovascular bundle lies between the internal and innermost layers, and the supply is from the intercostal nerves.

E Transversus thoracis muscle

The transversus thoracis pulls the costal cartilages downward, so it is an expiratory muscle. The diaphragm and external intercostals are the primary muscles of quiet inspiration, assisted by the levatores costarum and serratus posterior superior.

B Larynx of the airway

The larynx sits in the neck, above the thoracic inlet, and continues as the trachea, which does pass through. The left common carotid artery, recurrent laryngeal nerve, thoracic duct, and phrenic nerve all traverse the inlet.

C Seven pairs of ribs

Ribs one to seven are true (vertebrosternal) ribs, joining the sternum directly through their own costal cartilages. Ribs eight to ten are false ribs joining the cartilage above, and ribs eleven and twelve are floating.

C Transverse process of the fourth

The tubercle of a typical rib articulates with the transverse process of the numerically corresponding vertebra, so the fourth rib tubercle meets the transverse process of the fourth thoracic vertebra. (The head articulates with the bodies of that vertebra and the one above.)

A Superior intercostal artery

The first two posterior intercostal arteries come from the superior intercostal artery, a branch of the costocervical trunk from the subclavian artery. Only the third to eleventh come directly from the thoracic aorta.

B Thoracic aorta of chest

The third through eleventh posterior intercostal arteries (which includes the sixth to tenth) arise directly from the descending thoracic aorta, with the twelfth pair being the subcostal arteries.

D The transverse process of the same level

The tubercle articulates with the transverse process of the vertebra of the same number, a costovertebral joint. This is distinct from the head, which meets the vertebral bodies at the costovertebral joint.

A Internal thoracic and intercostal

At about one centimetre lateral to the sternal border, the internal thoracic artery runs vertically down, and at each space it gives anterior intercostal branches. A wound three centimetres from the midline at the fifth cartilage, two centimetres deep, catches both. The musculophrenic artery only begins at the sixth cartilage.

D Transversus thoracis muscle

The transversus thoracis depresses the ribs (expiration). The levatores costarum, serratus posterior superior, external intercostals, and the interchondral part of the internal intercostals all elevate the ribs.

C The beginning of the ascending aorta

The ascending aorta begins at the aortic valve, at about the third costal cartilage level, well below the sternal angle. At the sternal angle (disc between the fourth and fifth thoracic vertebrae) you find the tracheal bifurcation, the aortic arch beginning and ending, and the azygos vein entering the superior vena cava.

D Transversus thoracis muscle

The transversus thoracis lies on the inner surface of the anterior chest wall, arising from the posterior sternum and inserting onto the costal cartilages of ribs two to six. It is seen from inside at the lateral sternal border.

B Fifth to ninth thoracic

The manubrium lies opposite the third and fourth thoracic vertebrae. The sternal angle (manubriosternal joint) is at the disc between the fourth and fifth. The body runs from there down to the xiphisternal joint, which lies at the ninth thoracic vertebra. So the body of the sternum spans roughly the fifth through the ninth thoracic vertebral levels.

D Subclavian artery

The subclavian artery arches over the first rib and passes through the thoracic inlet between the anterior and middle scalene muscles. It is the vessel at risk with an injury at that level.

B Posterior intercostal artery

At the eighth intercostal space, the anterior part of the wall is supplied by the musculophrenic artery, but the wound in an intercostal space in the lateral or posterior wall reaches the posterior intercostal artery running in the costal groove of the rib above, together with the vein and nerve.

B Transversus thoracis muscle

The transversus thoracis lies on the internal (posterior) surface of the anterior chest wall and attaches to the costal cartilages of ribs two to six near the sternum. The other muscles listed are in or on the outer layers.

B Upward rotation of scapula

The serratus anterior protracts the scapula and, acting with the trapezius, rotates it upward so the glenoid faces up, allowing full overhead abduction of the arm. Its paralysis (long thoracic nerve) causes winging.

08 ANTERIOR ABDOMINAL WALL AND HERNIA

LECTURE 9

B The transversalis fascia is the posterior wall

This is the incorrect statement as written, because the transversalis fascia forms only the **lateral** part of the posterior wall; the medial part is reinforced by the conjoint tendon, which is precisely why direct hernias push through the weaker medial area. The other four statements are all true: the inguinal ligament is the in-rolled lower border of the external oblique aponeurosis; the lacunar ligament forms the sharp medial border of the femoral ring; the conjoint tendon is formed by the internal oblique and transversus abdominis aponeuroses; and the round ligament of the uterus is the female content passing through the superficial ring.

B Indirect inguinal hernia

A patent processus vaginalis leaves an open peritoneal track through the deep ring and along the canal. Bowel follows it, giving an indirect hernia that lies within the spermatic cord and lateral to the inferior epigastric vessels. This is the congenital type.

B Seventh and eighth thoracic

A subcostal (Kocher) incision runs about two to three centimetres below and parallel to the right costal margin, cutting across the dermatomes and the segmental nerves of the seventh and eighth thoracic levels.

D Superficial and inferior epigastric join in the sheath

The epigastric vessels form an important collateral network around the rectus abdominis, linking the upper body supply to the lower body supply — this can enlarge dramatically in inferior vena cava obstruction (caput medusae pattern). The other statements are all false: the **external** oblique aponeurosis forms the inguinal ligament; below the arcuate line (around the third or fourth lumbar level) the aponeuroses all pass **anterior** to the rectus, leaving no posterior sheath; the pyramidalis is supplied by the **subcostal** nerve is actually correct in some texts but the classic wrong statement here concerns the internal oblique, which inserts on the **lower three or four ribs** and the conjoint tendon, not simply ribs ten to twelve.

C Twelfth thoracic nerve

At two centimetres below and medial to the umbilicus, the dermatome is roughly the tenth or eleventh thoracic, not the twelfth (subcostal), which lies much lower, near the pubic region. A three centimetre stab there would pass the anterior sheath, the rectus, the inferior epigastric vessels behind it, and the posterior sheath (still present above the arcuate line).

D Deep circumflex iliac vessel

The deep circumflex iliac vessels run along the iliac crest, far laterally and inferiorly — nowhere near the umbilicus. A cut at the umbilicus would reach the anterior sheath, the rectus, the tenth thoracic nerve (the umbilical dermatome), and the posterior sheath.

E Direct hernia sac lies lateral to the artery

A direct hernia lies **medial** to the inferior epigastric vessels, pushing through Hesselbach's triangle. It is the indirect hernia that enters lateral to the vessels through the deep ring. The other four statements are correct.

E Internal spermatic fascia comes from the muscle

The internal spermatic fascia is derived from the **transversalis fascia**, not from the transversus abdominis muscle. The transversus abdominis contributes nothing to the coverings (it arches over the cord). The other statements are correct.

A It passes through the whole inguinal canal

An indirect hernia enters at the deep ring (lateral to the inferior epigastric vessels), traverses the whole length of the canal, and exits the superficial ring within the coverings of the spermatic cord. It is congenital and commoner in the young.

C It is due to weakness of inguinal triangle

A direct hernia is acquired, from weakness of the posterior wall in the inguinal (Hesselbach's) triangle, bounded by the inferior epigastric vessels, the rectus border, and the inguinal ligament. It bulges forward medial to the vessels and does not traverse the whole canal.

A Genitofemoral nerve

The genital branch of the genitofemoral nerve (first and second lumbar) supplies the cremaster muscle — the motor limb of the reflex. The sensory limb is the ilioinguinal nerve on the upper medial thigh.

E Inferior epigastric artery comes from femoral

The inferior epigastric artery arises from the **external iliac artery**, just above the inguinal ligament, not from the femoral artery. It is the landmark that separates direct from indirect hernias. The other statements are correct.

B Deep circumflex iliac artery

The deep circumflex iliac artery runs along the iliac crest, laterally — a suprapubic (Pfannenstiel) incision in the midline above the pubis stays well medial to it. That incision does cross the linea alba, the pyramidalis, the arcuate line region, and risks the inferior epigastric vessels laterally.

A Superficial inguinal node

Skin below the umbilicus and above the pubis drains to the superficial inguinal nodes, which then pass to the external iliac and para-aortic chains. The watershed is the umbilical level: above it drains to the axilla.

A Processus vaginalis

A patent processus vaginalis that stays open allows peritoneal fluid to track down and collect around the testis or within the cord, giving a communicating hydrocele. The same defect underlies the indirect hernia, so they often coexist.

D The external oblique aponeurosis makes the floor

The floor (inferior wall) of the inguinal canal is the in-rolled **inguinal ligament** with the lacunar ligament medially. The external oblique aponeurosis forms the **anterior wall** (and it is the source of the ligament, but it is not itself the floor). The other statements are correct.

A Superficial inguinal node

McBurney's point lies in the right lower quadrant, below the umbilical watershed line, so skin lymph drains to the superficial inguinal nodes. Deep structures such as the appendix itself drain to the ileocolic and para-aortic nodes.

C External spermatic fascia

The three coverings mirror the three flat muscle layers: the external oblique aponeurosis gives the external spermatic fascia at the superficial ring, the internal oblique gives the cremasteric muscle and fascia, and the transversalis fascia gives the internal spermatic fascia at the deep ring.

E Medial to pubic tubercle and below ligament

Note carefully: the standard clinical teaching is that a femoral hernia emerges **below and lateral** to the pubic tubercle, while an inguinal hernia emerges **above and medial** to it — this is the classic bedside distinction. Among the options offered in the source paper, the intended key is the one placing the hernia **below the inguinal ligament**, which is the reliable part of the distinction: the femoral canal lies deep to the inguinal ligament, so the sac always appears below it, in the upper thigh at the saphenous opening.

Check the wording. The standard bedside rule is that a femoral hernia appears **below and lateral** to the pubic tubercle, an inguinal hernia above and medial. The original option set does not express this cleanly; the reliable half of the distinction — below the inguinal ligament — is what is being tested here.

E Direct hernia passes medial to epigastric vessel

A direct hernia pushes through the weak posterior wall medial to the inferior epigastric vessels, in Hesselbach's triangle. The indirect hernia enters the deep ring lateral to the vessels and passes the whole canal. Each other statement swaps the two.

E Uterine artery of pelvis

A Pfannenstiel incision is made low and transversely, through skin, the anterior rectus sheath, and then the rectus muscles are separated in the midline. The uterine artery lies deep in the pelvis at the base of the broad ligament, far below the abdominal wall, so it is not encountered and is spared. Everything else listed is genuinely at risk: the inferior epigastric vessels run up the lateral edge of the rectus and are the classic bleeding source if the incision extends too far laterally; the iliohypogastric and ilioinguinal nerves cross the lower wall and entrapment is a well-known cause of chronic post-Caesarean pain; and the round ligament passes through the inguinal canal near the field.

QUICK REFERENCE

Answer grid

For fast marking

NO.	ANS	NO.	ANS	NO.	ANS	NO.	ANS	NO.	ANS
1	C	28	A	55	B	82	B	109	B
2	C	29	A	56	D	83	D	110	B
3	E	30	E	57	D	84	C	111	B
4	A	31	E	58	E	85	C	112	B
5	A	32	A	59	B	86	B	113	B
6	A	33	D	60	C	87	B	114	D
7	B	34	E	61	D	88	A	115	C
8	C	35	C	62	A	89	B	116	D
9	D	36	E	63	C	90	C	117	E
10	B	37	E	64	A	91	D	118	E
11	A	38	E	65	A	92	D	119	A
12	D	39	B	66	B	93	A	120	C
13	D	40	A	67	E	94	C	121	A
14	E	41	A	68	C	95	E	122	E
15	D	42	B	69	E	96	B	123	B
16	A	43	C	70	E	97	C	124	A
17	E	44	B	71	C	98	C	125	A
18	A	45	D	72	B	99	A	126	D
19	B	46	D	73	C	100	B	127	A
20	A	47	B	74	E	101	D	128	C
21	A	48	E	75	E	102	A	129	E
22	E	49	E	76	C	103	D	130	E
23	D	50	A	77	D	104	C	131	E
24	A	51	C	78	E	105	D		
25	D	52	D	79	C	106	B		
26	D	53	D	80	C	107	D		
27	C	54	B	81	C	108	B		

Distribution of correct answers — **A** 27 · **B** 26 · **C** 26 · **D** 26 · **E** 26 · total **131**. Spread deliberately even, so guessing by position gains you nothing.